

Interstellar exploration: From science fiction to actual technology

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ABSTRACT

The technology for even the most advanced missions in the solar system doesn't need advances in basic science. Traveling through the solar system can be described through what is called 'hard science fiction', i.e. science fiction strictly based on scientific knowledge. Interstellar exploration is a completely different matter. Robotic flyby missions to the nearest stars using nanoprobes can be performed using technologies based on known science, while anything beyond this requires advances which we don't know how to implement, or even we are not sure whether they are possible at all. The point applies not only to the technological aspects but even the scientific bases on which the relevant technologies may rest. The missions requiring less scientific-technological advances, are slow missions, like space arks (generation ships) or missions based on hibernation with travel times up to hundred years. To implement both, the uncertainties are more related to the advances in space medicine and biology than in propulsion and physics. The fastest travels allowed by the current interpretations of the relativity theory are relativistic missions in which the time contraction at speeds closing the speed of light is exploited to decrease the travel time for the astronauts, although the travel time seen by those who remain on Earth is close, in years, to the distance traveled expressed in light years. However, the energy required for this type of missions is large and grows drastically with the increase of time contraction.

Faster than light travel, which seems to be possible following some interpretations of relativity involving either wormholes or some sort of warp drive, requires substantial advances in fundamental physics. A symptom of this is that the novels dealing with interstellar travels belong more to the space opera – which doesn't follow strict scientific credibility – than to the hard science fiction subgenre. No novel of this kind explains details about how the relevant machinery works, and even less scientifically realistic are the movies, TV series, and video-games of this kind. Moreover, to achieve a travel time allowing to reach distant star systems in reasonable times using warp drives, the authors of Star Trek had to resort to the Warp Factor which is essentially a nonlinear scale. This makes the requirements for FTL travel even more difficult to achieve.

1. Introduction

The exploration of the solar system has already been started: robotic probes reached all the planets and many minor bodies, completely changing our understanding of the planets, satellites, asteroids and comets. Landers have reached not only Mars and Venus but also Titan and astronauts landed on the Moon and the plans to land humans on Mars are well developed. The technologies for even the most advanced missions in the solar system, like those being made possible by nuclear fusion propulsion, are being developed and even in these extreme cases – which include traveling to Mars in slightly more than one month – we need only to apply the knowledge science had produced in the last two centuries.

It is true that there are still debates on whether the most advanced

interplanetary propulsion devices we can imagine, namely propellantless drives like the much controversial EM-drive [2,3], are compatible with some basic principles of science, but such devices are not really necessary to explore and colonize the solar system, and we are not even sure whether they could outperform more conventional and not controversial devices like fusion drives [4].

If we want to imagine adventures in the solar system, we can do it through what is called 'Hard Science Fiction' (HSF), i.e. science fiction strictly based on scientific knowledge.

Interstellar exploration is completely different. Robotic missions to the nearest stars can be performed using technologies based on known science, in particular if we use nanoprobes travelling at a speed of 10–20 % of the speed of light, and we aim to perform a flyby of the target star or at most to enter in orbit about it [5]. Even these basic missions need the development of new hardware and software, based on still to be

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ACRONYMS

AI	Artificial Intelligence
BPP	Breakthrough Propulsion Physics
FTL	Faster Than Light
HSF	Hard Science Fiction
SF	Science Fiction
SRM	Sample Return missions
TRL	Technology Readiness Level
ZPE	Zero Point Energy

developed technologies. A sample return mission is much more problematic [6,7].

Note that this is not just for propulsion, but also for control – which requires levels of artificial intelligence not only we do not know how to implement, but even we are not sure whether they are possible at all – telecommunications and many other aspects.

Human missions (and colonization requires human missions) are even more critical.

Here the point is not only the technological aspects (technologies with very low TRL, below 3) but even the scientific bases on which the relevant technologies may rest (TRL <1).

There is no doubt that there are not many interesting plots which may be conceived describing robotic missions in which no human (or alien) is directly involved.

2. Types of interstellar missions

We could attempt a subdivision of robotic and human interstellar missions into a number of classes.

• Robotic missions

- R1 Perform a flyby crossing at high speed the destination system without manoeuvring in the system to get close to any planet.
- R2 Perform a flyby of one (or more) of the planets (or asteroids) by manoeuvring in the destination system.
- R3 Enter the system, achieving an orbit around the star(s) to study the system for a (potentially long) time.
- R4 Enter the system, achieving an orbit around one of the planets studying it for a (potentially long) time.
- R5 Landing on one of the bodies in the destination system if it has a surface. The lander may release a rover.
- R6 Landing on one of the bodies in the destination system, collecting samples, and then send them to Earth (SRM mission).
- R7 Landing on one of the bodies (possibly on an asteroid) in the destination system, where, apart for performing any of the tasks above, the probe builds a new probe (perhaps a replica of itself) which leaves the planet or the destination system to explore other planets or other systems. Probes of this kind are often referred to as Von Neumann probes [8–10]. These probes must produce a copy of themselves, and this is intrinsically dangerous: since this process may occur at a great distance in both space and time, from the present-time Earth, it will be impossible (perhaps except for the first few replications) to check that the replication process was carried on without errors, which can be considered as *mutations* [11].
- R8 Landing on a habitable planet carrying to it human life in latent form (e.g., frozen human embryos) and then give way to a human colony once at destination.

• Human missions:

- H1 Slow (i.e., lasting for one to several centuries) one way mission of the space-ark type. The starship must reach a speed from 0.1 % to 10 % of the speed of light. This kind of mission is mostly intended

for colonization purposes and assumes that a large crew is carried on board. The size of the crew is in this case independent from the travel time and is linked with the possibility of starting a healthy colony. On the contrary, the size of the starship (which is at any rate large) depends on the travel time, increasing with the journey duration. The main technological problems are linked with very long-lasting life support systems.

- H2 Slow (i.e., lasting for one to several centuries) one way mission, with hibernated crew. The spacecraft may be not very different from the previous one, except for a much smaller size. The main scientific and technological problems are linked with development of hibernation techniques, allowing the hibernated crew to survive for very long periods of time.
- H3 Fast (i.e., lasting a fraction of human lifespan) one-way mission. Even if the spacecraft can travel at about 10 %–50 % the speed of light, the journey time would be of several years, anywhere from 8 to 40, depending on the distance of the destination star from Earth, but only if it is very close. With these travel times, the situation is not much different from what seen for slow interstellar journey, and either the spacecraft is a space ark, or hibernation is required. Practically, the only advantage is that the hibernation or the life support system doesn't need to operate for decades or centuries.
- H4 Very fast (i.e., lasting a few years, but travelling at a speed at which relativistic effects are absent or weak) one-way mission. The starship must travel at about 50 %–80 % the speed of light. It is possible only for a few very close star systems because, even in this case, the travel time is several years, more than the time required for the slowest plans to reach Mars.
- H5 Relativistic one-way mission. The speed of the spacecraft is close to the speed of light and the travel time, in years (as seen by a fixed observer), is slightly longer than the distance in light years. However, due to the relativistic effects, the on-board time flows slower, and the travel time seen by the travellers is shorter. The travel time tends to zero with the speed tending to the speed of light, while the energy required for the travel tending to infinity. Colonization journeys are possible, and subjectively may be even very fast, but return journeys are practically impossible since those who come back do so when on Earth many years have passed. As Warren Salomon wrote in his *The economics of Interstellar Commerce*, the only reason one would embark in a relativistic return interstellar journey might be to get the interests of his money left in the bank [12].
- H6 Faster Than Light (FTL) one-way missions. Following present day physics, they are outrightly impossible and substantial advances (changes) in fundamental physics are required. At present the idea at the basis of FTL travel is the possibility of using a distortion of spacetime so that the distance to be covered by the spacecraft is shorter than the actual distance between the starting and arrival points.
- H7 Faster Than Light (FTL) two ways missions. Not only do they require the same substantial advances (changes) in fundamental physics as general FTL travel, but they also will be possible only if the apparent velocity is very high (by orders of magnitude) with respect to the speed of light.

The mission categories, which require less scientific-technological advances, are robotic missions, in particular fast flyby missions (R1 and R2), which are however characterized by a low scientific return, followed by one-way robotic missions in which the probe remains in the destination star system, perhaps landing on a planet or asteroid (R3 to R5). Sample return missions (R6) are much more demanding and require large advances, at least in the propulsion system.

The last two types of robotic missions (R7, R8), are even more complex requiring huge advances in Artificial Intelligence (AI) and biology (for R8 only), but above all involve also remarkable ethical

problems. A strong quarrel was triggered by the possibility of using Von Neuman machines (or better, probes; see for instance Refs. [11,13,14]). Von Neuman machines are a frequent topic of science fiction, starting from Arthur C. Clarke to Theodore Sturgeon and Greg Benford. It is worth mentioning the long series by Fred Saberhagen written from 1967 [15] to 2005. A more recent series on similar topics was written more recently by the author [16,17].

In these novels the Von Neumann machines, provided with either strong or weak AI, were built by aliens as automatic war machines or simply scientific probes which transformed in dangerous life destroyers by random mutations and natural selection. This sort of Darwinian evolution of machines was predicted just after Darwin wrote his book on evolution by Samuel Butler [18], and remains the main objection against the use of these type of probes [14].

Even stronger ethical problems are raised against the use of robotic probes (perhaps Von Neumann probes) to carry latent human life to extrasolar planets (R8) for colonization purposes. Owing to these objections we will not deal with this approach any more.

The only human interstellar missions which may be performed with present technology, or better with a technology developed in a predictable future, are slow missions, like space arks (generation ships) (H1) or missions based on hibernation (H2): in both cases the travel time may be hundred years.

To implement space arks we need to overcome two basic technological problems.

- Developing regenerative life support systems able to work for perhaps hundreds years with limited maintenance and without requiring spare parts and consumables from Earth
- Building huge spacecraft in an affordable way. This easily requires to build it on the Moon or on an asteroid using local materials

Both these requirements in turn require that humankind has reached the status of a spacefaring civilization. This is however not a serious problem because it is natural that before embarking into interstellar human travel our species needs to have explored and perhaps colonized a substantial part of our solar system.

Starship carrying hibernated humans are easier from this point of view, but require advances in AI since all maintenance and navigation tasks must be entrusted to machine.

To implement slow interstellar missions the uncertainties are more related to the advances in space medicine and biology than in propulsion and physics: for the former we are not sure that humans can withstand the psychological stress of a generation ship and the effects of radiation after exposure for generations, while for the second case the very feasibility of human hibernation (and above all of hibernation lasting centuries) has not yet been demonstrated.

Science fiction often described space arcs with or without hibernated crews, and an example worth mentioning is the movie *Passengers* (2016) directed by Morten Tyldum. Since the focus of this movie is more centered on the psychologic reactions of the passengers than on the technology, it belongs more to the space opera genre than to hard SF.

In fast interstellar missions (H3) the difficulties are more shifted toward the propulsion devices than to the size of the ship or the medical and psychological aspects, but this only if we aim to a star close to the Sun. If the star is several light years away the travel duration is anyway decades or centuries and the same problems seen for space arks and hibernation ship are present, together with stricter requirements for the propulsion system.

An example of a fast interstellar mission is the novel *Rocheworld* by Bob Forward [19] in which a one way mission to the Barnard's star (the second closest star) lasting 20 years is described. It is a scientifically well thought hard SF book, with the propulsion system described in some detail.

What we defined as very fast interstellar missions (H4) are similar to the previous ones, and at any rate last from some years to decades or

centuries. They do not differ much from the previous ones.

The fastest travel which is allowed by the current interpretations of the relativity theory are relativistic missions in which the time contraction at speeds close to the speed of light is exploited to decrease the travel time for the astronauts, although the travel time seen by those who remain on Earth is close, in years, to the distance travelled expressed in light years.

In a way, these missions do not differ much from slow missions with hibernated crew, with the difference that relativistic time contraction is used instead hibernation to avoid the aging of the travellers. The duration of the journey for those who do not travel remains of some years, for the closest stars, to many years of decades (or centuries) for more distant targets.

However, the energy required for this type of travel is very large and grows drastically with the increase of time contraction. If there is no problem regarding the theoretical bases on which such technology can be based, no propulsion system today available or even imaginable can allow such interstellar travels.

In science fiction it is possible to resort to the extraction of energy from vacuum (Zero Point Energy, ZPE). Again, descriptions of how this might be done are beyond present possibilities and also the novels developed in this way belong more to the Space Opera genre than HSF.

3. FTL travel

In most of science fiction dealing with interstellar travel, starships travel at FTL speed. This is mostly for narrative requirements, because only in this way many typical topics of science fiction, like two ways interstellar travel, interstellar commerce or war, two-ways contact with aliens, etc. are possible.

Even if it seems that FTL travel might be after all possible, following some interpretations of relativity involving either wormholes or warp drive, it would require such substantial advances in fundamental physics, that the novels, written by science fiction writers who dealt with interstellar travels, are more space operas than HSF. The authors make little attempt to describe how this is achieved with strict scientific credibility.

No written novels of this kind explain in some detail how the relevant machinery works, in particular how the huge quantities of energy required are obtained. In Star Trek it is said clearly that the warp drive is powered by the matter-antimatter reaction, the known reaction with the highest energy density. In particular, the deuterium-antideuterium reaction is mentioned.

But antimatter is just an energy vector and must be produced, which requires even more energy than the energy produced by the matter antimatter reaction. This may be accomplished by building a number of space stations which extract hydrogen (and deuterium, tritium or helium 3) from gas giants and use fusion reactions to produce the antimatter to refuel starships. This, contrary to FTL itself, is compatible with our physical knowledge, but the required technologies are still to be developed. Their TRL is close to 1, which is anyway better than 0 as in case of FTL.

Even less scientifically realistic are the very successful movies and TV series of this kind. The script writers of the series *Stargate Atlantis*, for instance, resort to Zero Point Energy which is still less realistic (at least with present day scientific technology) than antimatter generators.

Things are even less scientifically realistic in the many videogames, like, just to mention a few, the *Wing Commander* and then the *Privateer* series released in the 1990s by Origin Systems. The player is immersed in a quite involving simulation in which he has to interact with other (human and alien) characters and to pilot starships of different types. The whole simulation seems quite realistic and involve emotionally the player, who is led to identify with his/her character, even if fights look more like to dogfights between WWII aircraft than to what might be warfare in a distant future. Many other details are clearly unrealistic, perhaps by purpose – the user obviously has no experience of

hypothetical future space flight and thus considers realistic what he has some experience about.

Incoming spacecraft make noise (in the vacuum of space!), asteroid fields are dense beyond any possible realism and the player can navigate through them using his senses and reactions, just to name a few situations. But similar things occur also in the whole Star Wars saga and even Star Trek falls short from being really scientifically realistic.

One way (mostly colonization) FTL travel (H6) requires apparent velocities spanning from about 4 times the speed of light to reach the nearest star in one year (which anyway is more than traveling to Mars with present technology) to much higher speeds if the destination star is more distant.

Even higher velocities are required for two ways interstellar travel (H7), which is needed to achieve what is the mainstay of SF (or better, space operas): interstellar commerce, diplomacy (and perhaps war), crimes and intelligence (when space operas merge with detective or spy stories), romantic relationships, etc (Fig. 1).

To achieve a travel time allowing to reach distant star systems in a reasonable time, when we say faster than light we mean so much faster that it is difficult to even imagine it. Of this, the authors of Star Trek were well aware and they had to resort to the Warp Factor which is essentially nonlinear scale. Gene Roddenberry defined initially the warp factor as

$$w_f = \sqrt[3]{\frac{v}{c}}, \quad (1)$$

where v is the spacecraft speed and c is the speed of light.

This means that at a warp factor of 6 the spacecraft travels at 216 times the speed of light, reaching Alpha Centauri in 7.32 days. To look things in a greater environment, this speed is equivalent to about 0.18 parsec/day. To reach the Star Trek's planet Vulcan (identified as a planet orbiting star 40 Eridani or Keid, at 16.26 light years from Earth) it would take about 1 month. The definition in equation (1) was later changed, with an exponent 3/10 instead of 1/3 up to $w_f = 9$ and then reaching asymptotically infinity with $w_f \rightarrow 10$.

This example means just that to make it possible what science fiction describes, not only spacecraft must travel in FTL conditions, but their speed must be more than 2 orders of magnitudes greater than light speed. Just travelling slightly faster than speed has little advantages with respect to what in science fiction is called 'subluminal' travel. True FTL travel is even more difficult to achieve.

And even in this case, the situation will be like world wide travel in the nineteenth century: very costly (beyond the possibility of almost all

people, except the very rich and the government officers) and slow (requiring weeks or months). In this situation very few people could travel, but empires spanning more than one continent, and international diplomacy were possible.

Up to now, FTL travel belongs only to science fiction, and writers essentially describe two ways of implementing it: warp drives and space-time tunnels or wormholes. However, understanding more about warp drive was one of the targets of the Breakthrough Propulsion Physics (BPP) programme [3], and several serious physicists (the first was Miguel Alcubierre in 1994 [20] worked on it (Fig. 2). He declared he had been inspired by the fictional warp drive, first mentioned, before Star Trek, by John W. Campbell in his *Islands of Space* in 1931 (and in book form in 1957 [21]).

It seems that recently some progress toward the realization of a warp drive has been made, but the feasibility of this concept is still to be demonstrated.

There are still two points worth mentioning about FTL traveling. Where the hypothetical warp drive may be engaged or where the space-time tunnel (or wormhole) is located? It is very likely that this location is outside the departure (and symmetrically the exit point, is located outside the arrival) star system. This adds the time required to exit and enter the two systems, traveling at 'sublight' – to use a term common in science fiction – speed. This may mean that one or several days must be added to the total travel time. This would be negligible in other types of interstellar travel, but in case of FTL with a high warp factor, this time may constitute an important time delay. On the opposite, in some science fiction novels the space-time tunnels are described as connecting directly two points of space which may be at any distance so that the starship going through it passes instantaneously from one to the other. This way of implementing FTL travel raises even larger theoretical issues, similar to those raised by time travel. Above all, this seems to postulate the existence of an absolute time, a thing that is strictly incompatible with relativity theory. Imagine that a traveler goes through one of such tunnels: what is the time at which the arrival in the destination planet occurs? And if the same traveller comes back to Earth, what is the time of return? Is the solution that the arrival in the Earth system is that of departure plus the time spent in the other system, which can be computed using relativity, really compatible with this theory?

An interstellar travel thus takes a time equal to that required to exit the starting system and to enter the arrival one, independently of the distance of the two systems.

In the movie *StarGate*, and the TV series derived from it, the tunnels connect two points located on the surfaces of two planets – or even pf the

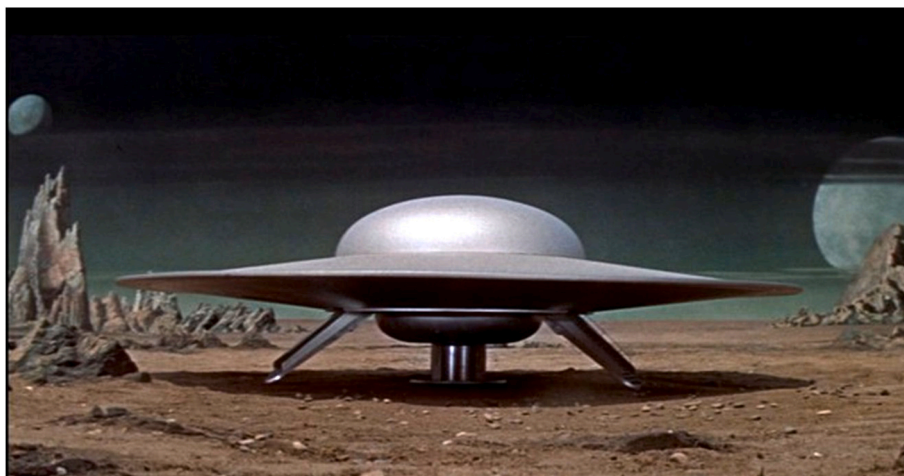


Fig. 1. The star cruiser C57D landed on Planet Altair 4 in the movie *Forbidden Planet* (1956). It was the first high-budget SF movie dealing with an interstellar journey performed at FTL speed. The original trailer recited: "Today, Man prepares to take his first steps outward into space – Tomorrow he will explore the stars. M.g.m's great technical staff brings you a magnificent picture of the distant tomorrow". Note that this was the year before the launch of the first satellite, Sputnik 1. It was a great success.

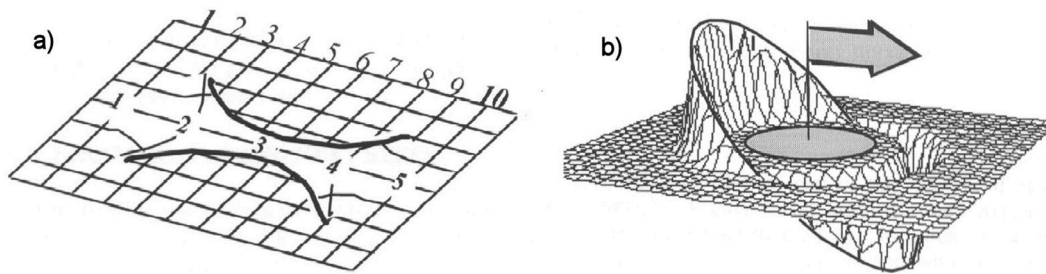


Fig. 2. Sketch of the space distortion as would occur in a) wormhole, and b) warp bubble. Note that in a) the apparent speed is less than twice the speed of light, so that the warp factor is less than 1.26. The images, from Ref. [1], were taken from papers of the BPP project.

same planet. In this way, the planet to planet journey occurs really in no time.

Even more interesting is what was described by Peter F. Hamilton (see for instance *Pandora's Star* [22] or *Judas Unchained* [23], belonging to his Commonwealth saga). The most unconventional feature is that tracks have been built to and from the 'stargates' (even if they are not called this way) so that interstellar travel is performed by train, with a rail system covering all the inhabited planets! Here the conceptual point of what happens during the crossing, when the train – and all object in it – are partially on a planet and partially on another (with each of them moving in space at different velocity, and thus with time passing at a different rate) is not explained.

The first application of this approach is described not without humor, with the first astronauts reaching, after a 'conventional' long journey in space, Mars and finding there two scientists who have preceded them by going through a space-time tunnel. This implies that space-time tunnels are discovered more or less at our time.

4. From science fiction to real technology

Up to now, we were mostly concerned with the problem of reaching speeds high enough to make interstellar travel possible. Both space scientists and SF writers are right in concentrating on this aspect, which is the enabling issue due to the huge interstellar distances.

However, to launch a successful interstellar mission, and above all a human interstellar mission, there are other aspects which need to be dealt with.

First, there are the human aspects, and, in a way, they may even influence robotic missions. If a robotic mission takes many tens of years to yield its scientific return (perhaps more than the working life of its initiators) the motivations to start it may be quite weak.

In case of human missions, the human factors become even more important, and the biggest one is perhaps radiation. Some sort of active shielding is required, to avoid the mass of the starship becoming enormous, with the exception of missions with hibernated crew. However, much research is at present being conducted on both technologies (active shielding and hibernation) and it is likely that they will be available in a reasonable time, to be applied also in interplanetary travel.

Microgravity can be easily solved, because a spacecraft designed to travel for decades or centuries like a space ark can be easily provided with artificial gravity (simulated by the rotation of the spaceship).

Another point is the acceleration: In slow interstellar travel the time required to accelerate, and then to decelerate in the arrival system, is a small fraction of the total travel time, so acceleration which can be easily withstood by the crew can be accepted. Accelerating at 1 g it takes 347 days (almost 1 year) to get close to the speed of light. This means that entering a wormhole at a speed close to the speed of light with an acceleration that can be withstood by the crew, would take several months.

In science fiction this problem is solved by introducing devices which are defined acceleration compensators (or inertial compensators or

inertial dampeners), that allow people to withstand arbitrarily high accelerations by manipulating gravity. They could also create an artificial gravitation field without rotating the spacecraft, and thus inducing also Coriolis accelerations. However, to create such devices we should be able to manipulate space time and inertia, a thing we not only don't know how to do, but even whether they are possible at all. The only thing we can say is that if technology will ever allow to create devices like warp drives, it would also create acceleration compensators.

Another difficulty will be dealing with the psychological problems of people living in a community travelling for years – or for all their life – in space. To this the answer may be hibernation.

The on-board production of all commodities people need – from food to medicines, from spare parts to all the consumables, and so on – is another problem. For this the experience of the lunar, Mars or space colonies will be precious.

Another point are the telecommunications, which however may lose importance with the growing distance of the space crew from the solar system. Already with Mars, the distance is large enough to make a true conversation problematic; when the time delay becomes months or years (and in interstellar travel the time delay can be more than one century) people may use telecommunications only to remain informed on what happened on Earth.

In science fiction this is solved mostly in two ways: FTL telecommunication or material messages.

In the first case, together with travelling at FTL speed, also FTL communication have been realized. In Star Trek 'subspace messages' travel at a speed higher than light speed, and may take weeks to travel between fairly distant star systems. In the development of the saga subspace communications became faster to the point of becoming instantaneous and allowing true conversations. Again the problem of absolute time becomes important but is neglected.

In the second case, the information is recorded on a material substrate, which is put on a small automatic starship, often defined a probe, and sent to destination at FTL speed. This is more realistic, in the sense that if FTL space travel has been achieved, there is no problem in sending messages in this way. The probe is much smaller than a starship, and is usually assumed to be faster so that it is worthwhile to send a probe with a recorded message instead of sending a true starship.

The situation reminds that of the 19th century, where long distance communications were as fast as a letter carried by the fastest horse (or carrier pigeon, and then train) or ship.

It is interesting to remember that in the 1960s the first experiments of re-enter from orbit were motivated by recovering the films with the images of Earth taken by the early spy satellites. The film was then developed on Earth. Even during the Apollo missions the pictures taken on film and then developed on Earth were of much better quality than those sent by radio (the Apollo 12 TV camera failure prevented from having any picture broadcast by that mission).

The suggestion of using a similar technique for the data coming from the outer parts of the solar system was forwarded in view of robotic missions to the Kuiper belt.

5. Conclusions

The distance separating the stars of our galaxy, at least in the zone where the Sun is, are so large that interstellar travel requires speeds which are well beyond our present capabilities. Since the energy required to reach a given speed is proportional to the square of the speed (in classical physics), interstellar travel requires enormous quantities of energy. The revolution in physics carried out by relativity for certain aspects makes things more difficult and for others simpler.

More difficult because when getting close to the speed of light the mass of a body increases and so the energy grows more, tending to infinity when the speed tends to the speed of light. Another point is that the above reasons make it impossible to travel at the speed of light so the minimum travel time in years between two stars is equal to (or better, larger than) their distance in light years.

Easier because, when getting close to the speed of light, the time on-board of the starship contracts and for the travellers the travel time is shorter than that recorded by those who remain at home. There is more: the energy required for reaching a speed, in which this time contraction occurs, is less than what would be the energy required to perform the same travel in classical physics in the same subjective time.

For the above reasons, and above all for the energy requirements, until a few decades ago many scientists thought that interstellar exploration, both robotic and human, was impossible. No wonder, if you think that in 1926 a British scientist, A.W. Bickerton, wrote: *This foolish idea of shooting at the moon is an example of the absurd length to which vicious specialisation will carry scientists. To escape Earth's gravitation a projectile needs a velocity of 7 miles per second. The thermal energy at this speed is 15 180 calories [per gram]. Hence the proposition appears to be basically impossible* [1]. Just 40 years later, Apollo 11 landed two human beings on the lunar surface. Note that the reason for which Bickerton said it was impossible is exactly the same that many people mention to state that interstellar travel is essentially impossible.

From Prehistory to the industrial revolution the quantity of energy available to humankind was unbelievably (for us) small. Then it increased quickly and continues to increase today, even if at a slower rate. The expansion of humankind into space will cause our energy needs to expand, but also will allow us to increase our energy availability to expand.

Travel in the solar system will be achieved fully only when nuclear fusion will be fully developed. And it is likely that interstellar travel will require the use of the matter-antimatter reaction, but this is feasible within the frame of present day theoretical scientific knowledge.

So we can say that slow interstellar travel, be it with space arks or by hibernating the crew, will become possible in due time. This will make it possible to colonize a large number of exoplanets. Interstellar travel using relativistic effects is also possible, but we can say that it will materialize later than slower travel, owing to the huge quantities of energy it requires.

Interstellar travel has been for decades one of the basic themes of science fiction – be it hard SF or space opera. However, the plot of most of the novels (or movies, TV series or videogames) requires two ways interstellar travels, performed in a time much shorter than what is possible by traveling at the speed of light.

Moreover, the speed required by those plots is so high that sometimes it is measured in a nonlinear scale (warp factor) and sometimes the interstellar travel is instantaneous. The possibility of realizing this type of interstellar travel in the real world is much more problematic.

However, there are at least two possible ways which perhaps might allow to implement FTL travel: warp drive and space-time tunnels or wormholes. Some scientists assert that they are not incompatible with general relativity, but on the contrary, they are possible solutions to the equations on which this theory is based.

Notwithstanding some statements assessing that some steps in this

direction have recently been performed, such an achievement requires advances in the understanding of general relativity, or more likely the unification of this theory with quantum mechanics in the so-called quantum relativity.

This, if possible at all, might occur in an undetermined future, and then it must be followed by the development of the ensuing technology, or better, technologies because FTL propulsion is not the only important aspect, but advances in a number of other technologies, which have little to do with propulsion in itself, are required.

Many of these technologies will anyway be developed to make human exploration and colonization of the solar system possible and later, to achieve slow interstellar travel [7], so it is likely that when (and if) FTL travel will be achieved, they will be ready.

For now, we can just enjoy tales about FTL journeys and proceed with the exploration of the solar system and the first attempts to go beyond it (for instance launching precursor interstellar probes) using realistic technologies.

CRedit authorship contribution statement

Giancarlo Genta: Conceptualization, Data curation, Formal analysis, Investigation, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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